

The Prime-Indicator Matrix Does Not Obey the Semicircle Law: Disproof of TLMC Conjecture 00000000142

TLMC submission (machine-verified: Lean 4, zero axioms)

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Abstract

TLMC Conjecture 00000000142 asserts that the empirical spectral distribution (ESD) of the prime-indicator matrix $M_n = (\mathbf{1}_{i+j \text{ prime}})_{1 \leq i, j \leq n}$ tends to the semicircle law, with a CLT for the linear spectral statistics. We refute the conjecture. The matrix is symmetric with 0/1 entries, whence two exact trace identities: $\text{tr } M_n = 1$ for all n , and $\text{tr } M_n^2 = N(n) := \#\{(i, j) \in [n]^2 : i + j \text{ prime}\}$. Since $N(n) \sim n^2 / \log(2n)$ (prime number theorem with partial summation), the ESD second moment is $m_2(n) = N(n)/n \sim n / \log(2n) \rightarrow \infty$, while the semicircle law has second moment 1; the conjecture's own LSS-CLT clause presupposes $m_2(n) \rightarrow 1$. Under the normalized reading, $\|M_n/n\|_{\text{op}} \leq \sqrt{N(n)}/n \rightarrow 0$, so the limit is δ_0 , not the semicircle. Numerically the raw spectrum spreads over the scale $\pm\sqrt{n/\log n}$ (RMS 7.2 at $n = 300$, 16.3 at $n = 2000$), so no tight limit exists. All finite- n identities used are machine-checked in bare Lean 4 with zero axioms.

1 The conjecture

Conjecture 1 (TLMC 00000000142). *The empirical spectral distribution of the matrix $M_n = (\mathbf{1}_{i+j \text{ prime}})_{1 \leq i, j \leq n}$ tends to the semicircle law, and the linear spectral statistics satisfy a CLT.*

Because every prime exceeding 2 is odd, $M_{ij} = 1$ forces $i + j$ odd (except the single entry $(1, 1)$): M_n is a bipartite (block off-diagonal) matrix up to the rank-one perturbation $e_{11}e_{11}^\top$, hence by Weyl's inequality its spectrum is symmetric about 0 up to a $1/n$ perturbation; all eigenvalues are real.

2 Exact trace identities

Lemma 1 (Trace one). *For every $n \geq 1$, $\text{tr } M_n = 1$.*

Proof. $\text{tr } M_n = \#\{i \leq n : 2i \text{ prime}\}$. If $i \geq 2$ then $2i$ is even and at least 4, hence composite; and $2 \cdot 1 = 2$ is prime. So the count is 1. $\square$

Lemma 2 (Second moment). *For every $n \geq 1$, $\text{tr } M_n^2 = N(n) := \#\{(i, j) \in [n]^2 : i + j \text{ prime}\}$.*

Proof. M is symmetric with entries in $\{0, 1\}$, so $\text{tr } M^2 = \sum_{i,j} M_{ij}M_{ji} = \sum_{i,j} M_{ij}^2 = \sum_{i,j} M_{ij} = N(n)$. $\square$

Proposition 1 (Asymptotics of $N(n)$). *$N(n) \sim n^2 / \log(2n)$ as $n \rightarrow \infty$. Consequently the ESD second moment $m_2(n) = \frac{1}{n} \text{tr } M_n^2 = N(n)/n \sim n / \log(2n) \rightarrow \infty$.*

Proof sketch. $N(n) = \sum_{p \leq 2n} g_n(p)$ with $g_n(p) = \min(p-1, 2n+1-p)$. Since $g_n(p) = n\varphi(p/n)$ with $\varphi(t) = \min(t, 2-t)$ a fixed continuous function on $[0, 2]$ and $\int_0^2 \varphi(t) dt = 1$, partial summation with the prime number theorem $\pi(x) \sim x/\log x$ gives $N(n) \sim n^2/\log(2n)$. (Convergence is logarithmically slow; numerically $N(n)\log(2n)/n^2 \approx 1.09\text{--}1.11$ for $300 \leq n \leq 8000$, drifting toward 1.) $\square$

3 Refutation

Theorem 1 (Raw reading). *If the ESD of M_n tends to the semicircle law and the linear spectral statistics satisfy a CLT in the standard (polynomial test function) sense, then in particular $\frac{1}{n} \text{tr} M_n^2 \rightarrow \int x^2 d\sigma_{\text{sc}}(x) = 1$. But $\frac{1}{n} \text{tr} M_n^2 = N(n)/n \rightarrow \infty$ by Lemmas 2 and Proposition 1. Contradiction.*

Theorem 2 (Normalized reading). $\|M_n/n\|_{\text{op}} \leq \|M_n\|_F/n = \sqrt{N(n)}/n \leq \sqrt{\pi(2n)}/n \rightarrow 0$ by the prime number theorem. Hence every eigenvalue of M_n/n lies in $[-\sqrt{2/\log(2n)}, \sqrt{2/\log(2n)}]$ eventually and the ESD of M_n/n converges to δ_0 , not to the semicircle law.

Remark 1 (Relation to the recorded attack). *The recorded verdict recomputed $\text{tr} M_n^2/n^2$: 0.172, 0.156, 0.145 at $n = 300, 600, 1000$ and concluded the limit is δ_0 . Our independent recomputation confirms those numbers exactly (0.1715, 0.1562, 0.1452). One normalization is corrected here: $\text{tr} M^2/n^2$ is the squared Frobenius density, not the ESD second moment, which is $\text{tr} M^2/n$ and diverges; the honest δ_0 statement is Theorem 2, for the eigenvalue-normalized matrix M_n/n . Under either normalization the conjecture is false — in the raw reading it fails more strongly than recorded, since no tight limit exists at all.*

Remark 2 (What carries the raw-reading refutation). *Weak limits are not determined by moments without uniform integrability; the divergence of $m_2(n)$ alone would leave the logical loophole of mass escaping to infinity. Here the conjecture's second clause closes it: a CLT for linear spectral statistics with polynomial f entails the centered moment convergence used in Theorem 1, and that fails outright. Independently, the numerics of Section 4 show the bulk itself (not a vanishing fraction) at scale $\sqrt{n/\log n}$, so no distributional limit of the ESD of M_n exists, tight or not.*

4 Numerical verification

Exact integer counts $N(n)$ (sieve; also fixed by `decide` in Lean for $n = 8, 64, 128$: 23, 941, 3239):

n	$N(n)$	$N(n)/n^2$	$1/\log(2n)$	$N(n)/n$ (ESD m_2)	$n/\log(2n)$
300	15439	0.1715	0.1563	51.46	46.90
600	56237	0.1562	0.1410	93.73	84.63
1000	145171	0.1452	0.1316	145.17	131.56
2000	528537	0.1321	0.1206	264.27	241.14
4000	1944355	0.1215	0.1113	486.09	445.08
8000	7187475	0.1123	0.1033	898.43	826.42

Spectral statistics (symmetric eigensolver; `reproduce.py --eigen`):

n	$\lambda_{\max}$	$\lambda_{\min}$	RMS	$\frac{1}{n}\#\{ \lambda \leq 1\}$	$\frac{1}{n}\#\{\lambda < 0\}$	$\frac{1}{n}\#\{ \lambda > 3\}$
300	51.72	-51.71	7.17	0.1200	0.5000	0.5767
600	94.03	-94.03	9.68	0.0667	0.5000	0.7400
1000	145.62	-145.61	12.05	0.0520	0.5000	0.7860
2000	264.96	-264.95	16.26	0.0320	0.5000	0.8690

Semicircle reference values: second moment 1; $\sigma_{\text{sc}}([-1, 1]) = \frac{1}{3} + \frac{\sqrt{3}}{2\pi} \approx 0.609$; $\sigma_{\text{sc}}(\mathbb{R} \setminus [-3, 3]) = 0$. The negative fraction is exactly 1/2 at all n (bipartite symmetry), the only statistic that agrees. Note $\lambda_{\max} \approx N(n)/n$: the Rayleigh lower bound $\lambda_{\max} \geq \mathbf{1}^\top M_n \mathbf{1}/n = N(n)/n$ is nearly tight, so the outlier pair $\pm \lambda_1$ alone accounts for the bulk of $\text{tr } M_n^2$.

5 Boundary: proven, heuristic, open

Proven and machine-checked (Lean 4, zero axioms): Lemma 1 and Lemma 2 for all n ; the exact values $N(8) = 23$, $N(64) = 941$, $N(128) = 3239$; $N(n) > n$ at $n = 64, 128$ (ESD second moment above the semicircle’s total second moment); the growth witness $N(128)/128 > N(64)/64$; $\text{tr } M_{1000} = 1$ by kernel computation.

Proven at PNT level (cited, not formalized): $N(n) \sim n^2/\log(2n)$; $m_2(n) \rightarrow \infty$; $\|M_n/n\|_{\text{op}} \rightarrow 0$ and $\text{ESD}(M_n/n) \rightarrow \delta_0$.

Heuristic (stated as such): the bulk of the spectrum of M_n spreads over a symmetrized quarter-circle / Marchenko–Pastur type shape at scale $\approx \sqrt{2n/\log(2n)}$ (the bipartite block A has density $\approx 2/\log(2n)$ and the rank-one mean contribution escapes as the outlier pair $\pm n/\log(2n)$).

Open: a rigorous identification of the rescaled bulk limit; a proof that a *positive fraction* of eigenvalues escapes any fixed compact (Frobenius counting gives only $\gtrsim c \log n$ eigenvalues outside $[-K, K]$); linear spectral statistics of any rescaled model.

6 Reproduction

```
python3 reproduce.py          # exact counts, trace identities, verdict comparison
python3 reproduce.py --eigen  # + eigenvalue statistics (numpy)
cd lean4 && lake build && lake env lean Check.lean
```

7 Machine verification

Bare Lean 4 (toolchain `leanprover/lean4:v4.33.1`, no Mathlib, `lakefile.toml` with `defaultTargets = ["Main"]`). `Check.lean` prints `#print axioms` for all 18 theorems; every line reads `does not depend on any axioms` (no `sorryAx`, no `propext`, no `Quot.sound`, no `Classical.choice`). The proofs use only `rfl`, `decide` (kernel computation), explicit `Eq.trans/congrArg/Eq.subst` term proofs and structural induction; `omega`, `simp` and `native_decide` are deliberately avoided because they would introduce axioms.

References

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